

Fabrication Techniques of Superhydrophobic Coatings: A Comprehensive Review

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Superhydrophobic coatings are growing in demand in textile, wearable electronics, and healthcare devices due to their nonwetting, antibacterial, anti-icing, and self-cleaning properties. With potential applications of superhydrophobic surfaces in consumer electronic devices, there is a huge demand to develop new materials and fabrication techniques that can help make high-performance, scalable electronic devices directly onto the conductive and flexible substrates for a wide range of applications. Superhydrophobic materials which are highly stretchable and exhibit good electrical conductivity find applications in wearable electronics sensors, energy storage devices, anticorrosion circuits, etc. Superhydrophobicity can be achieved by creating micro/nanostructures using materials with low surface energy. In general, most of the superhydrophobic coatings are developed by making the materials' surfaces rough by multistep procedures. However, due to the lack of excellent natural templates as well as simple and cost-effective manufacturing procedures, a coating approach that is simple, cost-effective, scalable, and environmentally friendly is in high demand. Finding appropriate materials, simple fabrication techniques, and cost-effective, durable, and eco-friendly superhydrophobic coatings for real-life applications is still challenging. This article reviews the materials used, fabrication techniques, the theoretical background of hydrophobicity in surfaces, and the limitations of existing methods. It also reviews the superhydrophobic responses of different materials.

1. Introduction

A superhydrophobic surface is one that has a water contact angle (WCA) of more than 150° ; if the surface is slightly tilted, the droplet will roll off or slide away.^[1] Due to the self-cleaning properties of superhydrophobic surfaces, these surfaces demonstrate a high potential for “dust-phobicity” and minimal dust adherence. Creating superhydrophobic surfaces that inhibit dust collection is a current research topic.^[2] Many superhydrophobic surfaces are found in nature, such as lotus leaves, butterfly wings, desert beetle scales, red rose petals, mosquito eyes, moth eyes, gecko feet, and fish scales.^[3] The secretion of wax on the rough surface

of lotus leaf allows deposited water droplets on the leaf to simply roll off and naturally clean the lotus leaf. These natural unique surfaces have given rise to a novel concept for creating artificial superhydrophobic materials. Hydrophobic coatings are in great demand due to their range of applications, namely, cooking utensils, clothing, windows of tall buildings, antibacterial, eyeglasses, etc. Superhydrophobic surfaces have some unique properties like corrosion protection, antireflection, durability in terms of hardness, scratch resistance and adhesion, self-cleaning, self-healing, etc.^[4–6]

The wettability of a surface is referred in terms of contact angle of a liquid drop on the surface. Larger is the angle, lesser will be the contact with the surface. This implies that a superhydrophobic surface has a greater WCA ($>150^\circ$) and small water sliding angle ($<10^\circ$).^[7] A water droplet that lies on a superhydrophobic surface has a form of a perfect sphere due to which it easily rolls off. A superhydrophobic surface has low surface energy and greater surface roughness which makes the water droplet settles on the rough surface where air gets

trapped. The trapping of air on rough surface introduces an increase in surface hydrophobicity.^[8]

These superhydrophobic surfaces are particularly interesting because of antisticking, anticontamination, and self-cleaning qualities. Anti-sticking of snow for windows and antennas, antibiofouling paints for boats, metal refining, stain resistant fabrics, self-cleaning windshields for vehicles, and antisoiling architectural coatings are all examples of applications where these features are desirable. A surface's wettability is proportional to its surface energy. Low surface energy materials, such as a surface with hexagonally packed $-CF_3$ groups, can have WCAs as high as 120° .^[9]

The scientific and engineering value of superhydrophobic surface is dramatically limited as it has weak mechanical abrasion resistance. As mentioned above, although different superhydrophobic surfaces have been constructed by various techniques, few coatings were reported to be actually used in industrial applications because of their poor mechanical abrasion resistance and surface chemical stability. Some superhydrophobic surfaces are even fragile to the finger contact. Obviously, the acquiring of excellent mechanical abrasion resistance and chemical stability for superhydrophobic surface is an urgent demand for their practical applications.^[10]

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Superhydrophobic and superoleophobic surfaces and coating technologies that are simple, quick, efficient, and low cost are widely sought. The limited mechanical and chemical resilience of produced superhydrophobic and superoleophobic surfaces significantly limits their use. After repeated cycles of usage or exposure to severe conditions, many surfaces quickly lose their superhydrophobicity and superoleophobicity. Designing more durable superhydrophobic surfaces is an unmet need. The fundamental factors of stability, such as substrate materials, surface shape, and surface chemistry, must be thoroughly investigated. The key development in this discipline is superhydrophobic and superoleophobic surfaces with numerous functionalities and smart features.^[11]

Supramolecular chemistry has been widely employed to make a variety of functional polymers, including healable materials. Because of their potential applications in flexible electronics, wearable devices, and soft robotics, silicone-based polymers with a healable function have gotten a lot of attention. The damage-healable slippery and oil-repellent surface are two examples of applying supramolecular silicone for coating development. Damage-healing and superhydrophobicity in silicone-based supramolecular nanocomposites have also been demonstrated recently. However, as bulk materials like foam and aerogel, these supramolecular nanocomposites are handled and assessed. Processibility, substrate binding, and damage repair as a coating on solid substrates are all things that remain elusive.^[12] With the increasing growth and widespread demand for superhydrophobic surfaces, developing a universal superhydrophobic coating that can be employed on both hard and soft substrates and can withstand massive shape deformations without losing its antiwetting properties remains a problem.^[13]

Many studies have been conducted to establish a correlation between the factors responsible for superhydrophobicity of the surfaces. Nanoscale roughness, surface energy, coating thickness, mechanical stability, and substrate-coating binding strength are some of the parameters.^[4,14] Different models have been employed, such as Wenzel's and Cassie's, to correlate and comprehend the link between surface roughness and superhydrophobicity. Furthermore, several methods and techniques have been employed to change the substrate surface to a superhydrophobic surface with adjustable qualities. Some of the techniques used are UV technique, photopolymerization, and spray technique.^[14] The focus of this study is on the fabrication processes and procedures employed for the fabrication of superhydrophobic surfaces and also the materials used to develop superhydrophobic coatings. In addition, many uses of superhydrophobic coatings are addressed in this article.

1.1. Nanomaterials in Development of Superhydrophobic Coating

The study of hybrid material nanocomposites materials science pays close attention to the numerous benefits in the creation of sustainable nanostructured surfaces, as well as their numerous applications in the sectors of healthcare, energy, and environment.^[15,16] Designing of superhydrophobic coatings dispersed with nanoparticles is drawing attention of researchers across

the world. Generation of micro- and nanolevel roughness and lowering the surface energy solves the problem of dust accumulation easily. This technology inhibits the adherence of the solid surface to water droplets while favoring the attachment of water droplets and dust particles, allowing the surface to self-clean.^[2] Nano- and microhierarchical roughness and low surface energy of the solid substrate are two important criteria to consider when designing a superhydrophobic surface. Because nanomaterials typically have at least one dimension in the region of 1–100 nm, nanostructures are an appealing option for depositing their layer on solid surfaces to create nanoscale roughness. Because of the nanoscale size and high specific surface area to volume ratio, nanostructures offer unique physical and chemical characteristics.^[2]

Mohamed S. Selim et al.^[17] synthesized low-cost SiO₂-ZnO/PDMS nanocomposite for superhydrophobic application. Author reported that microroughness present on the surface, increase in the contact angle ($167^\circ \pm 2^\circ$), and decrease in the surface energy (9.24 mN m^{-1}) indicate the fouling release (FR) and self-cleaning properties of nanocomposites. Author reported the PDMS/ γ -Al₂O₃ nanorod composite for FR application. Author reported that for 0.5% of nanofillers the dispersion of filler in polymer matrix is homogenous and it gives the maximum CA ($169^\circ \pm 1^\circ$) and minimum surface free energy (10.03 mJ m^{-2}) for self-cleaning and FR application. Further increasing in nanofiller concentration 0.7% and 1% results in the decrement in CA, i.e., $143^\circ \pm 2^\circ$ and $122^\circ \pm 2^\circ$, respectively.^[18] Author anchored the surface of alumina nanorod by using the treatment of graphene oxide and prepares the hybrid GO- γ -Al₂O₃ nanorod (average diameter 30–40 nm). Up to the 1 wt% of nanofiller gives the good dispersion in polydimethylsiloxane (PDMS) matrix and maximum contact angle 151° with decrease in surface free energy up to 13.25 .^[4] Author also fabricated the PDMS-based nanocomposite by reinforcing the graphene oxide anchored zinc oxide nanorod (average length $<1 \mu\text{m}$ and diameter 30–40 nm) and titanium oxide (average length 20 nm and average diameter 10 nm). With the inclusion of 1 wt% of hybrid ZnO nanofiller, resulting materials show superhydrophobic behavior with WCA of 151° and WCA of 152° for inclusion of hybrid TiO₂ nanorod with 1 wt% concentration.^[19,20]

MnO₂ is a promising candidate among possible metal oxides due to its low cost, easy handling, high activity, oxidation property eco-friendly nature, etc. Using the controlled synthesis parameter of nanofiller (average diameter 20–30 nm, β -phase [100] direction) and distribution of nanofiller in silicon matrix, the β -MnO₂/silicon nanocomposite is fabricated for 1 wt% of filler concentration. This fabricated nanocomposite gives higher value for WCA 158° and decrease in surface energy value to 12.65 mN m^{-1} .^[15]

Incorporation of nanoparticle in polymer matrix becomes the trending research for enhancing the superhydrophobic nature of polymer. Different nanoparticles like metal oxide, semiconductor, hybrid nanoparticles, 1D, 2D, 3D, etc. play the different role to tune the properties of materials. Along with the different techniques of fabrication materials are gaining an equal importance in designing the superhydrophobic films and coating materials.

1.2. Global Trends in Research of Superhydrophobic Coating

The superhydrophobic market is predicted to reach USD 51.18 million by 2026, according to Mordor Intelligence, with a CAGR of 25.48% from 2021 to 2026.^[21] Recent advances in underwater electronics, as well as flexible and wearable electronics, are projected to present future business prospects. Superhydrophobic coatings for food packaging are expected to be in high demand in the future for a few applications. Due to rising demand for novel fabrics and high-end footwear, the textile and footwear industries are likely to dominate the market and rise over the projection period.^[21]

The research articles published from 2000 to 2021 (Scopus database) yearly is shown in **Figure 1a**. This rising trend in research of superhydrophobic coating correlates the growing market in this area.

Primary keyword used to access this published data form Scopus database is “Superhydrophobic.” Superhydrophobic coating is used in many areas like anti-icing, self-cleaning, anticorrosion, and antiwetting and also in industries like textile, footwear, buildings and construction, etc. (Figure 1b). These few keywords are used as secondary keywords to get the insight of publications in this area.

The information retrieved from Scopus database reveals that a lot of research is going on for fabricating superhydrophobic coatings for various applications across the world. Considering the growing interest and demand of superhydrophobic coatings, there is a strong need to improvise the available materials, methods, and applications and discover new materials and fabrication techniques.

2. Theoretical Background

Surface wetting happens when water spreads over the surface as a result of the presence of high-energy molecules with a greater affinity for water on the surface. Water droplets have a low static contact angle on these surfaces, whereas water droplets on superhydrophobic surfaces have a high static contact angle, repelling water drops. As the WCA rises, the droplet becomes more spherical and makes less contact with the surface.^[22]

Based on the contact angle, a surface's wettability is categorized into four categories, such as 1) “superhydrophilic

($\theta \approx 0^\circ$, complete wetting of the surface),” 2) “hydrophilic ($\theta < 90^\circ$, partial wetting of the surface),” 3) “hydrophobic ($\theta > 90^\circ$ but $< 150^\circ$, partial nonwetting of the surface),” and 4) “superhydrophobic ($\theta > 150^\circ$, complete nonwetting of the surface).”

Sliding angle hysteresis (SA) (α) or contact angle, also known as contact angle hysteresis, is determined as the difference between the advance (θ_A) and retreating contact angles (θ_R) created when the surface is tilted (**Figure 2a**). The gravitational attraction causes the liquid droplet to start rolling down. The sliding angle of the surface must be $< 10^\circ$ for a superhydrophobic sample.^[22–24]

In superhydrophobic surfaces, surface roughness is extremely essential. Water droplet behavior can be affected by roughness.

Because of the pinning of the droplet in between the grooves, the sliding angle increases. According to the Wenzel theory, wettability rises for hydrophilic surfaces as roughness increases, while hydrophobic surfaces have the poorest wettability.

The Cassie–Baxter model describes a rough surface as one that is uneven in character and has air gaps between the grooves. The droplet shatters and does not travel through the surface. When compared to the Wenzel model, the sliding angle or contact angle hysteresis is negligible in this scenario.^[22–24]

Various wetting states of liquid with solid surface are given in **Figure 2b**.^[22,26] Water spreadability on a solid surface is known as wettability. Surface energy determines the contact angle of liquid with solid surface on a flat surface (**Figure 2b-i**). The contact angle is calculated using Young's equation as follows^[26]

$$\cos(\theta) = \frac{\beta_{SA} - \beta_{SL}}{\beta_{LA}} \quad (1)$$

Where, θ , and β_{SA} , β_{SL} , β_{LA} are representing the contact angle of smooth surface and interfacial tension for the solid–air, solid–liquid, and liquid–air respectively.

In 1940, Wenzel and Cassie modified Young's formula (Equation (1)) to assess the contact angle quantitatively for rough surfaces. According to the Wenzel model, the liquid droplet completely fills the tiny grooves of the solid rough surface (**Figure 2b-ii**) because of which actual contact between liquid and solid surface increases. As the contact angle is increased, the hydrophobic effect is geometrically magnified. This model is used to compute the contact angle of a droplet on a

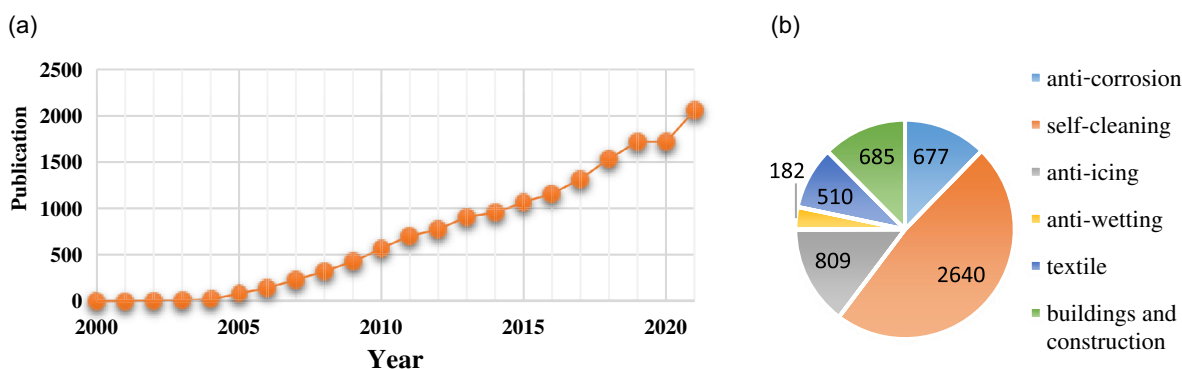


Figure 1. Superhydrophobic coating: a) trends in publication; b) publication related to different applications. Source: <https://www.scopus.com/> (accessed on 20/01/2022).

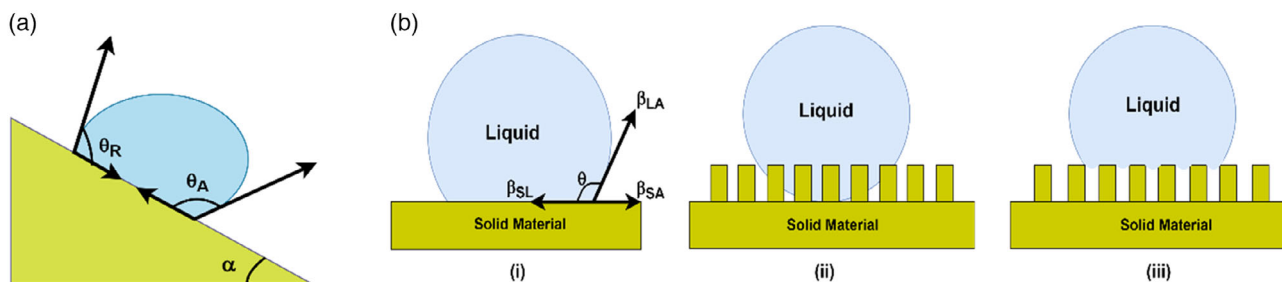


Figure 2. a) Water droplet on an inclined plane;^[25] b) water droplet wetting states on a solid surface: i) Young's, ii) Wenzel, and iii) Cassie.^[22,26]

homogeneous surface.^[26] The Wenzel expressed the contact angle as in Equation (2)

$$\cos(\theta_W) = r \cos(\theta) \quad (2)$$

where θ is the Young's contact angle (intrinsic contact angle), θ_W is the Wenzel's contact angle (apparent contact angle), and r is the roughness factor. The roughness factor (r) is the proportion of the actual surface area to the projected area (planar surface area).^[22,26]

For porous and more roughed surfaces, the absolute value for the right-hand side of Wenzel's equation may be greater than 1. When this happens, the Wenzel model breaks down, and the Cassie–Baxter model takes over.^[22] Cassie and Baxter (Figure 2b-iii) updated the preceding idea by considering the proportion of each phase (components) contact area. This model is based on the assumption that 1) a liquid droplet rides on top of roughness properties and rests on a composite air–solid surface; 2) a heterogeneous surface is composed of many materials to average the free surface energy.^[22] If a composite surface is made up of two separate components, θ_1, f_1 and θ_2, f_2 are the intrinsic contact angle and apparent fraction of each component, respectively, and $f_1 + f_2 = 1$. The Cassie–Baxter equation is used to calculate the apparent contact angle for composite materials

$$\cos(\theta_{CB}) = f_1 \cos(\theta_1) + f_2 \cos(\theta_2) \quad (3)$$

The liquid drop only contacts the tip of the peaks in this model, while air packets inside the groves prevent the drop from touching the surface. The apparent contact angle in Equation (3) is adjusted and provided in a modified form (Equation 4) because the intrinsic contact angle between air and droplet is 180° .

$$\cos(\theta_{CB}) = f_{SL} \cos(\theta) + f_{SL} - 1 \quad (4)$$

Superhydrophobic coatings are prepared with composite materials. Artificial superhydrophobic coatings can be designed by inclusion of nanoparticles of silica, zinc oxide, Fe_2O_3 , halloysite clay, etc. in the polymer matrix. The parameters affecting superhydrophobicity of the surface are static WCA, surface energy, abrasion resistance, mechanical robustness, thickness of the coating, bond between substrate and coating, effect of temperature and moisture on coating, etc. To modify the substrate surface, certain methods such as self-assembly, surface coating, chemical vapor deposition, and spray technique are employed to obtain superhydrophobic surfaces. Various polymer matrix with outstanding features are considered to be good matrix materials

for designing an optimum superhydrophobic polymer nanocoatings.^[27]

3. Development in Superhydrophobic Coating

This section discusses the applications, materials used, and the methodology used in the development of superhydrophobic coatings.

Das et al.^[14] discussed ZnO dispersed polymer nanocoatings used for oil/water separation, clay-based polymer nanocomposites used for antiwetting and self-cleaning properties, CNT. He observed that silica-based polymer nanocomposites showed extreme chemical resistance properties. His work also explained the recent developments, fabrication methods, and applications of superhydrophobic surfaces in various fields. Nguyen-Tri et al.^[28] reported the fundamentals of wettability and surface tension phenomena. The study emphasized the importance of roughness in the formation of superhydrophobic surface. In order to protect solar cells, transparent superhydrophobic coatings with high stability are high in demand.

Xue et al.^[29] introduced the use of superhydrophobic surfaces in commercial products. His review focused on important commercial applications of superhydrophobic surfaces and substrates for large areas. Self-assembly method was used to fabricate multilayer ionic assembly of polyelectrolyte, which created few layered grafts, and was found to be robust and steady. Zhang et al.^[30] reported basic theories explaining superhydrophobicity and the recent growth in this field. This study also reported anti-corrosion mechanisms of superhydrophobic surfaces. By decreasing water contact region and time or by introducing additional air barrier films, the metal substrates and aqueous corrosive bodies were made to communicate and interact less with each other.

Milonis et al.^[31] presented an outline of growth and development based on mechanical robustness of superhydrophobic surfaces. They also discussed mechanical wear and assessment techniques. Fihri et al.^[32] focused on superhydrophobic coatings to protect steel from corrosion. Natural superhydrophobic coatings, wettability of a rough surface using different models, and various equations involving static contact angle were discussed in his work. Different methods of fabrication of synthetic nonwettable coatings with effect on anticorrosivity property were also studied. Lathe et al.^[33] investigated different methods to create artificial superhydrophobic surfaces. The author reported the

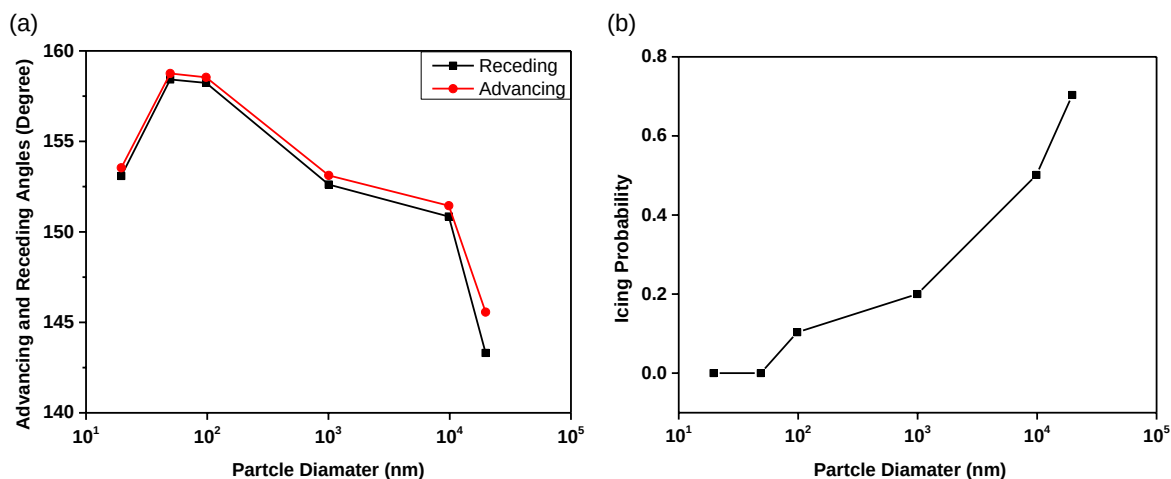


Figure 3. Effect of particle size on a) advancing and receding angle; b) icing probability. Adapted with permission.^[34] Copyright 2009, American Chemical Society.

recent development in synthesis, characterization, theoretical modeling, and their applications.

Cao et al.^[34] demonstrated the anti-icing property of superhydrophobic surfaces fabricated by electrodeposition method using particle-dependent polymer nanocomposites. The authors reported that some of the properties of the superhydrophobic surface changed due to variability in particle size. The effect on WCA and icing effect due to change of particle size is shown in Figure 3.

Kang et al.^[35] fabricated nanosilica surface covered with epoxy embedded in epoxy matrix giving epoxy resin nanosilica composite coatings, i.e., reinforcing nanosilica filler, which included organic molecules containing hydrophilic group such as radical epoxy in the form of desired functionality that reinforced with EP matrix. Adding ample quantities of hydrophilic silica to composite epoxy resin coatings indicated less friction coefficient and lowered break rate than that of epoxy coating, the study reported.

Su et al.^[36] developed superhydrophobic surfaces by dispersing nanosilica in epoxy coating. Owing to the ruggedness induced by the incorporation of more nanosilica, the fixed contact angle and the receding contact angle of the samples were increased (Table 1). The coatings were found to be fully transparent to visible light (Figure 4).

Varshney et al.^[37] reported synthesis of steel mesh surface by chemical etching method to form rough microstructures.

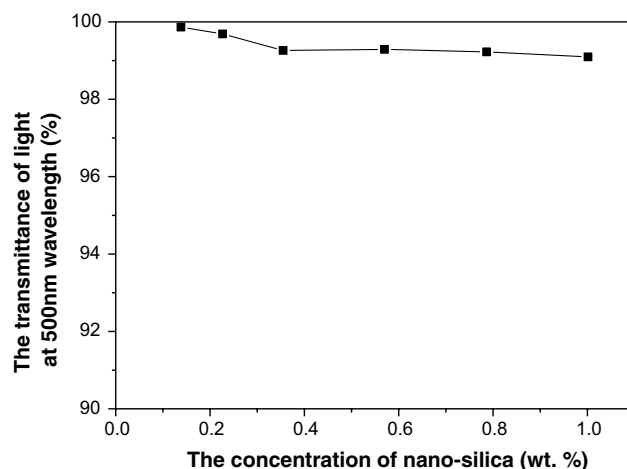


Figure 4. Transmittance of light through coating for different concentration of nanosilica Adapted with permission.^[36] Copyright 2006, Elsevier B.V.

Durable and regenerable superhydrophobic and superoleophilic steel mesh surfaces for separating oil from water were designed by chemical etching method.

Nanda et al.^[38] reported chemical etching technique where a mixture of iron chloride and hydrogen chloride was treated with

Table 1. The contact angle on surface of sample. Adapted) with permission.^[36] copyright 2006, Elsevier B.V.

Concentration of nanosilica [wt%]	Static angle [°]	Advancing angle [°]	Receding angle [°]	Hysteresis of contact angle [°]	Sliding angle [°] for 10 μ L water droplet
0	105	108	32	76	>30
0.1	110	114	40	64	>30
0.25	117	121	47	64	>30
0.5	147	158	106	52	>30
0.75	160	162	117	45	>30
1.0	168	170	151	19	11.8

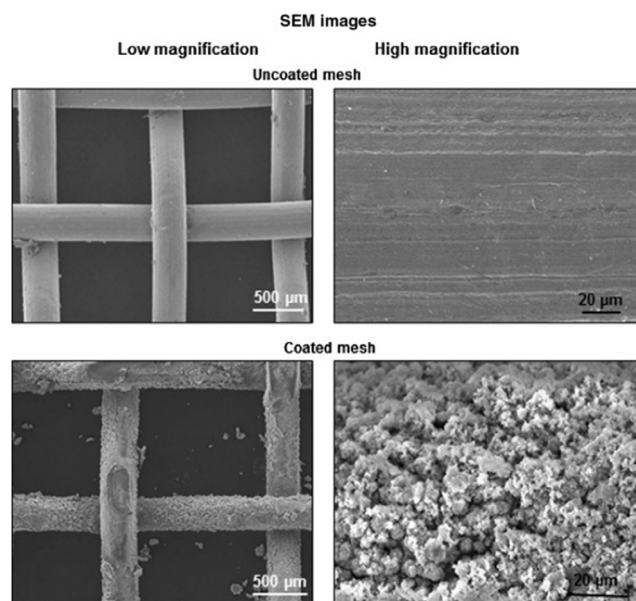


Figure 5. SEM image of chemically etched surface of stainless steel. Adapted with permission.^[38] Copyright 2018, Elsevier Inc.

hexadecyltrimethoxysilane. Flower-like microstructures on the surface of the stainless steel mesh were created (**Figure 5**) and WCA was determined (167°). Excellent thermal, chemical, and mechanical stability of the coatings was also observed.

Using potassium hydroxide and lauric acid, Varshney et al.^[39] fabricated superhydrophobic coatings on Al metal substrates. Water contact angle for the fabricated coating found to be $>150^\circ$. These coatings exhibited exceptional self-cleaning and anticorrosion behavior. There was an elevated contact angle due to increase in immersion time. Water jet effect, water surface flotation, and low-temperature condensation tests confirmed the water-repellent quality of the coatings.

Lomga et al.^[40] fabricated superhydrophobic coatings on aluminum using potassium hydroxide and lauric acid by chemical etching process. Rough microstructures on treated surfaces were seen to attribute to superhydrophobic nature of the coating. Contact angle increased as etchant concentration and etching time was increased (**Figure 6**). Coating superhydrophobicity remained intact after performing tape peeling test and water jet impact with 15 cycles. The coating showed no change on 90° and 180° bending, indicating good mechanical durability (**Figure 7**). Coating could also withstand 40° – 170°C temperature range and prolonged exposure to UV radiation.

Aditya Kumar et al.^[41] showed designing of superhydrophobic surfaces by the process of chemical etching using nitric and hydrochloric acid followed by hexadecyltrimethoxysilane treatment. Superhydrophobicity was observed with WCA of 162° and SA of 4° . Coating showed mechanical, chemical, and thermal stability. The coating was discovered to be self-cleaning and water-repellent, and rough microstructures with rectangular pits were seen while studying surface morphology.

Varshney et al.^[42] fabricated superhydrophobic aluminum surfaces through chemical etching method. Surface morphology

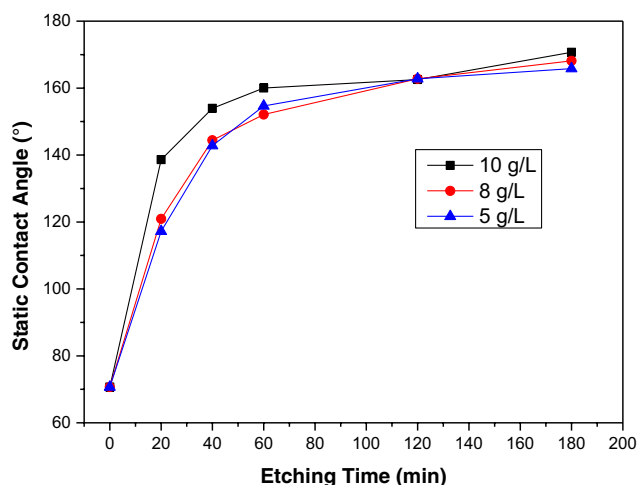


Figure 6. Contact angle variation of Al surface for different etching concentration and time. Adapted with permission.^[40] Copyright 2017, Elsevier B.V.

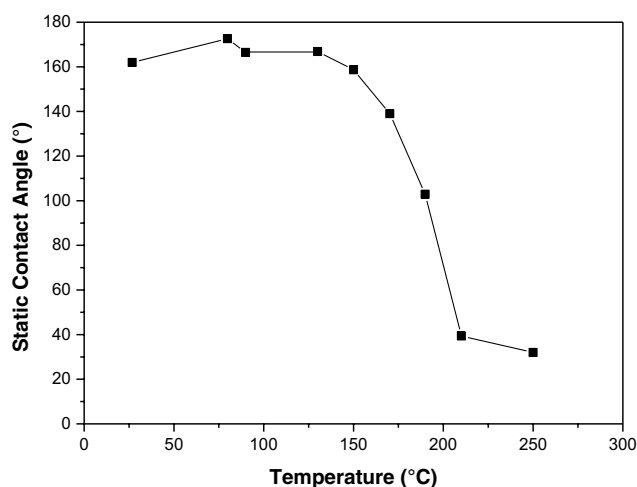


Figure 7. Contact angle variation of Al surface with different annealing temperature. Adapted with permission.^[40] Copyright 2017, Elsevier B.V.

indicated rough microstructures on the coating. Superhydrophobicity of the synthesized coating with WCA of 172° and SA of 4° was reported. Coating showed mechanical, chemical, thermal, and UV stability. Coating exhibited self-cleaning and antifogging nature and excellent regenerable property.

Liu et al.^[43] used an acid etching procedure to create superhydrophilic and superhydrophobic surfaces on an aluminum alloy substrate. The WCA is found to be 159° by coating PP-g-MAH polypropylene (PP) grafting maleic anhydride, while it was only 141° when coated with PS polystyrene. It was observed that the surface customized by silane pairing agent before PP coating resulted in improvement in stability and solvent opposition performance.

Aluminum alloys have various industrial applications such as automobile, aviation, and building construction. Degang Xie et al.^[44] synthesized aluminum superhydrophobic coating by

initially immersing and etching in boiling solution of NaOH for 5 min and then for 30 min in an ethanol solution of lauric acid. The contact angle was reported to be greater than 150° with a negligible hysteresis when the concentration of NaOH solution was greater than 5 g L^{-1} . This method is fast, low cost, and reliable for industrial applications. Liao et al.^[45] synthesized micro/nano dual-structured aluminum surface by continuous chemical etching. Cold water was able to agglomerate in big droplets on the fabricated aluminum surface and roll off before freezing. Such fabricated aluminum surface could also decrease the freezing area and ice accumulation in glazed ice. It also showed long-term mechanical stability.

Qi et al.^[46] observed on the crystalline silicon wafer; pyramidal hierarchical structures were formed by KOH etching and silver catalytic etching. This structured silicon wafer showed strong antireflection and superhydrophobicity after fluorination. UV-assisted imprint lithography technique was used to form the superhydrophobic substrate by dispersing the Si structure to the NOA 63 film.

Satapathy et al.^[47] presented superhydrophobic coatings of SiO_2 nanoparticles embedded in LLDPE on glass substrate which were fabricated by dip coating method. Porosity was varied using phase separation method, and the coating was characterized for self-cleaning, WCA, water repellency, and surface morphology, thermal, chemical, and mechanical properties. Subsequently, the superhydrophobic copolymer films were fabricated by means of the method of phase separation. The microstructures of polymer films were governed by the degree of phase separation, which was properly enhanced by the ethanol concentration. The variation of contact angle with ethanol concentration is shown in **Figure 8**. This fabricated coating shows WCA of 170° and SA of 3.8° . The author reports that this coating is useful for the application in self-cleaning technology, and coating required under perturbation conditions like chemical and thermal environment.^[47]

Bravo et al.^[48] illustrated superhydrophobic transparent films built using nanosilica particles using multilayer self-assembly

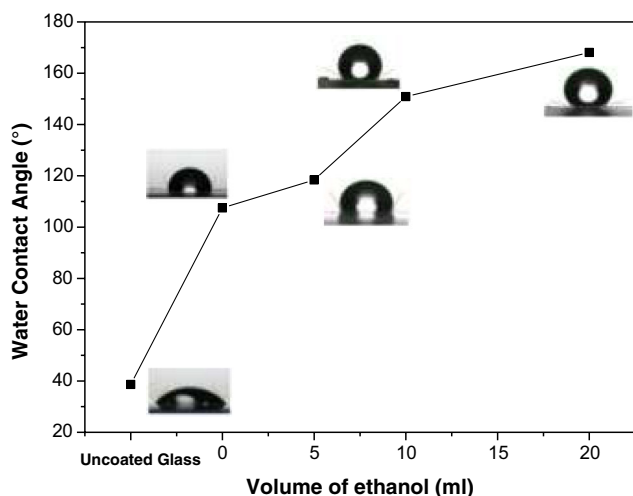


Figure 8. Contact angle variation with ethanol concentration in porous LLDPE. Adapted with permission.^[47] Copyright 2017, Elsevier B.V.

method. He claimed that if the position and stage of aggregation of nanosilica particles of various sizes were managed, then the resulting thin multilayer film could achieve optimum surface roughness with limited light dispersion. Nanoporous thin multilayer film was able to integrate effect of antireflection. **Figure 9** shows variation of advancing contact angle and contact angle hysteresis as a function of film roughness.

Yin et al.^[49] used solution immersion and surface self-assembly technique to construct biomimic superhydrophobic surface on copper substrate with $\text{WCA} > 150^\circ$. After treating these surfaces with NaOH and potassium persulfate solution, morphological study showed a nanosize needle structure containing irregular flowered accumulation. The nanoneedles were crystallized and hydrophilic surface could be modified in superhydrophobic surface by using dodecanoic acid and self-assembly method. Cu is widely used engineering material and this method of surface modification is easy way to form biomimic superhydrophobic surface on engineering metals.^[49]

Xu et al.^[50] produced superhydrophobic, highly crystal-clear coatings by sol-gel process. Customized hollow silica nanoparticles (HSNs) of APTS [3-aminopropyltriethoxysilane] were dip-coated on glass substrates. Residual solvent was extracted and thermal annealing process was then used to solidify the coatings. Chemical vapor deposition (CVD) technology was used for 1H,1H,2H,2H-perfluorooctyltriethoxysilane (POTS) coatings that demonstrated superhydrophobicity and high precision clarity. Simultaneous characteristics of superhydrophobicity and strong clarity were suggested with versatile concentration of APTS and withdrawing the speed of dip coating. This fabricated superhydrophobic coating showed $\text{WCA } 156^\circ$ with an $\text{SA} < 2^\circ$. This coating could be used for eyeglasses, windows of high-rise buildings, solar cell glass panels, windshields of vehicle, etc. in future by improving their mechanical stability.^[50]

Michele Manca et al.^[51] fabricated robust superhydrophobic and antireflective surfaces using sol-gel process. Solid surfaces with suitable unevenness and low surface energy were generated without any additional hydrophobic capping layer by injecting

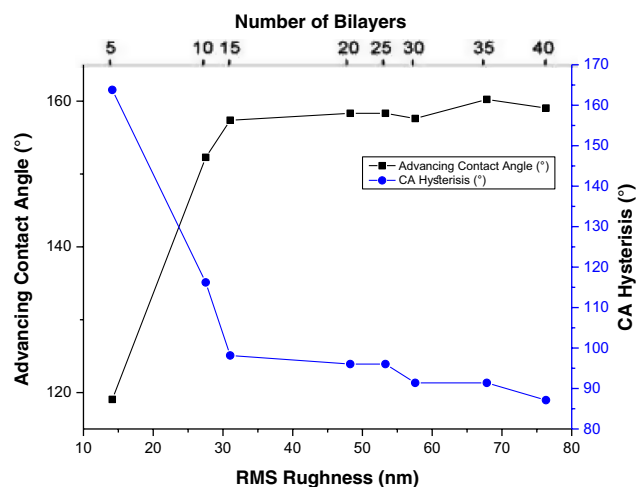


Figure 9. Advancing contact angle and contact angle variation with film roughness. Adapted with permission.^[47] Copyright 2007, American Chemical Society.

functionalized silica nanoparticles TMS (trimethylsiloxane) into an organo-silica gel matrix. This organo-silica gel matrix (sol C) was prepared by a trimethoxysilane (MTES) acid catalyzed hydrolysis and condensation. It was discovered that a sufficient proportion of hydrophobic nanoparticles and silica gel matrix is critical for achieving maximum water repellence and superhydrophobic surface thermomechanical stability. The upper nanoporous layer supported regular effects of antireflection.

Ribeiro et al.^[52] produced hybrid polymer films by dispersing silica nanoparticles (SiNPs) in the polymer matrix with increased tensile strength and impact resistance. Particle aggregations in the films were minimized and a homogeneous distribution of the inorganic components could be achieved. Hierarchically structured materials could be prepared by coating with particles from hybrid polymer. Sol-gel method was used to create these hybrid polymer films, where silica was deposited on the surface of the polymer matrix. This control spacing between silica particles in final material was useful in applications like photoactive composite.

Lakshmi et al.^[53] reported hydrophobic-modified coating in which colloidal zinc hydroxide (CZH) was integrated into the polyvinylidene fluoride (PVDF) sol-gel matrix to create superhydrophobic sol-gel composite film. CZH was prepared by limited precipitation and by dealing with an optimized concentration of stearic acid. Customized CZH film containing roughness was soaked in tetraethyl orthosilicate (TEOS) and MEOS acid-catalyzed soil and sprayed on spray-coated glass slides. This synthesized superhydrophobic coating shows WCA of 165° with an SA < 2°. Author reported that zinc hydroxide and stearic acid were nonhazardous, nontoxic, cheap, and easily available. This method of fabrication of superhydrophobic coating has potential in water-repellent textile and healthcare applications.^[53]

Suhao Wang et al.^[54] described the fabrication of superhydrophobic coatings. Polystyrene grains or polyethylene terephthalate grains were immersed in toluene and then nanoparticles of silica were dispersed for 30 min in the solution ultrasonically. The filter paper was immersed in this solution and dried in an oven to evaporate the solvent. Filter paper applications were investigated in sorting out liquids, for instance, oil and ethanol from water. The superhydrophobic filter paper showed good oil uptake ability. WCA of this modified filter paper for different liquid is shown in Figure 10.

A polymer-functionalized CNT film was developed by spray casting of one-step CNT polystyrene. The film demonstrated not only consistent superhydrophobic properties, but also good light transmission in the visible light spectrum, with a WCA of around 160° and a sliding angle of less than 3°.^[54]

Si et al.^[55] documented recent progress of superhydrophobic and superhydrophilic materials synthesized using biological methods. They also addressed emerging applications and advancements in this area and future research scope. This work introduced a remarkable 3D printing technology that could rapidly and accurately prepare the surface structure and morphology of certain superhydrophobic materials developed using biological background. The synergistic effect between hierarchical dual scale structure and intrinsic material characteristics is the most important aspect of bioinspired superwetting material design. Bioinspired superhydrophobic materials have enormous

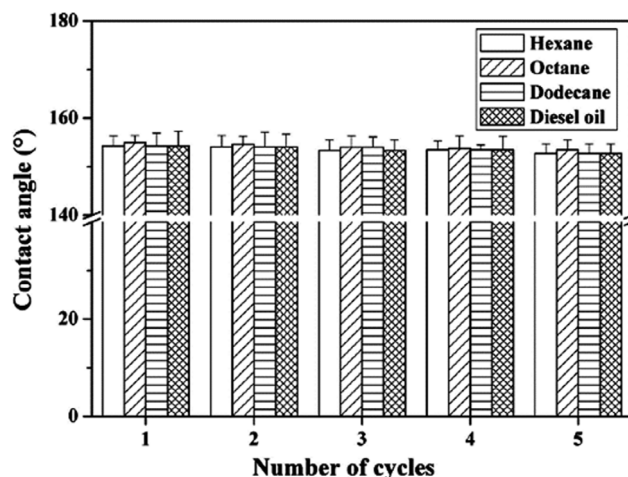


Figure 10. Water contact angle of modified filter paper in different oils. Adapted with permission.^[54] Copyright 2010, American Chemical Society.

application potential in fields such as self-cleaning, anti-icing, water repellence, corrosion resistance, drag reduction, and so on.^[55]

Simpson et al.^[5] discussed the basic characteristics of hydrophobicity and superhydrophobicity, as well as other fundamental characteristics of superhydrophobic surfaces, and their photolithographic growth. Superhydrophobic spray commercial availability was also investigated on these surface-efficient coatings but lacked durability and cost-effectiveness. The study also highlighted solid, long-lasting superhydrophobic coatings, paints, and surfaces with transparent superhydrophobic optical thin films.

Grari et al.^[56] developed and manufactured two different multilayer polypyrrole/silica composite coatings called electrophoretic deposition (EPD) by a highly efficient technique. The method of electrophoretic construction allowed for the extensive inclusion of silica particles in the coatings. The simple immersion in the coating of silica nanoparticles did not yield homogeneous repartition inside the coating. Dispersed silica particles made the coatings more anticorrosive. Furthermore, it was observed that when the silica layer was applied to the surface of stainless steel, the repartition of the particle was found to be more homogeneous.

4. Fabrication Methods for Synthesizing Superhydrophobic Surfaces

Superhydrophobic surfaces can be created using a variety of approaches. Some are simple and cost-effective, while others need multistep procedures, severe temperatures, or specific reagents and equipment, all of which raise the cost of coating.

4.1. Electrochemical Deposition Method

This method is also referred as electrodeposition method.^[26] The micro/nanostructured surface is created by electrochemical deposition for super hydrophobicity. The various types of

Table 2. Examples of superhydrophobic coatings by electrochemical deposition. pDop, poly(dopamine).

Matrix	Coatings	Electrolyte composition	Process conditions	WCA	SA/CAH	References
Copper sheet	Zn	$\text{NH}_3 + \text{KCl} + \text{Zn}(\text{CH}_3\text{COO})_2$	-1.35 V , 25 min	$170^\circ \pm 2^\circ$	$\approx 0^\circ$	[58]
Steel	Zn/pDop/NDM	$\text{ZnSO}_4 + \text{KCl} + \text{dopamine} + \text{N-dodecyl mercaptan (NDM)}$	1.5 V, 30 min	167.6°	$< 1^\circ$	[59]
Al	Zn/pDop/NDM	$\text{ZnSO}_4 + \text{KCl} + \text{dopamine} + \text{N-dodecyl mercaptan (NDM)}$	1.5 V, 30 min	166.5°	$< 1^\circ$	[59]
Cu	Zn/pDop/NDM	$\text{ZnSO}_4 + \text{KCl} + \text{dopamine} + \text{NDM}$	1.5 V, 30 min	164.9°	$< 1^\circ$	[59]
Gold	Fluorinated polypropylene	$\text{Bu}_4\text{NClO}_4 + \text{acetonitrile (anhydrous)} + \text{pyrene monomer}$	$-0.7 \text{ V to } E^w$ (1.4–1.9 V), scan rate 20 mV s	160°	1°	[60]

nanostructures in the form of nanodots, nanowires, and nano-flowers can be obtained by regulating electrolyte concentration and reaction times.^[34] This is a simple and repeatable method. Also, this method is less time-consuming, easy to process, controllable for desired morphology and thickness, and scalable with high efficiency.^[26] This procedure uses a solution containing ions or charged micro/nanoparticles to deposit a thin coating on a conductive substrate. Metal (and its compound) electrochemical deposition, nonmetallic electrochemical deposition, and composite electrochemical deposition are the three types of electrochemical deposition based on the deposition medium.^[57] Examples of electrodeposition superhydrophobic coating are given in **Table 2**. Schematic of electrodeposition is shown in **Figure 11a**. Su et al. created roughness on copper substrate using facile and low-cost electrodeposition process and then heat treatment is used for grafting the low surface energy material on it. This roughened surface has WCA of $162^\circ \pm 1^\circ$ and SA of $3 \pm 0.5^\circ$. The schematic shown by Su et al. for the evolution of surface deposited with Ni and then grafting of (heptadecafluoro-1,1,2,2-tetradecyl)triethoxysilane (AC-FAS) by heat treatment is redrawn and represented in **Figure 11b**.^[10]

4.1.1. Metal Electrodeposition

In this process, discharging of metal ions takes place at the interface of the solution and electrode under the influence of an electric field, resulting in the formation of a new metal phase.

In the electrodeposition process, first the ions are discharged at electrode when it transfers from electrolyte to the electrode, and then atoms are deposited in the lattice to form the crystal structure. The fundamental law of crystallization is altered by the electric field at the cathode contact, and the two processes interfere with each other. There are two types of processes: one in which the newly formed adsorption atoms come together to form crystal nuclei, which grow to form crystals over time, and another in which the newly formed adsorption atoms spread on the surface of the electrode, and then they enter the network to continue to grow on the original metal mesh.^[57]

4.1.2. Nonmetallic Electrodeposition

Many nonmetals, such as low surface energy conducting polymers, can achieve superhydrophobicity through one-step electrodeposition, obviating the requirement for additional surface modification and broadening the spectrum of anticorrosive coating applications.^[57] Carbon chains containing fluorine atoms have low surface energy and can be used to synthesize superhydrophobic coating combining them with suitable organic substances.^[57]

4.1.3. Composite Electrodeposition

This technique is used to get the properties of two different materials which are matrix material and filler material. The

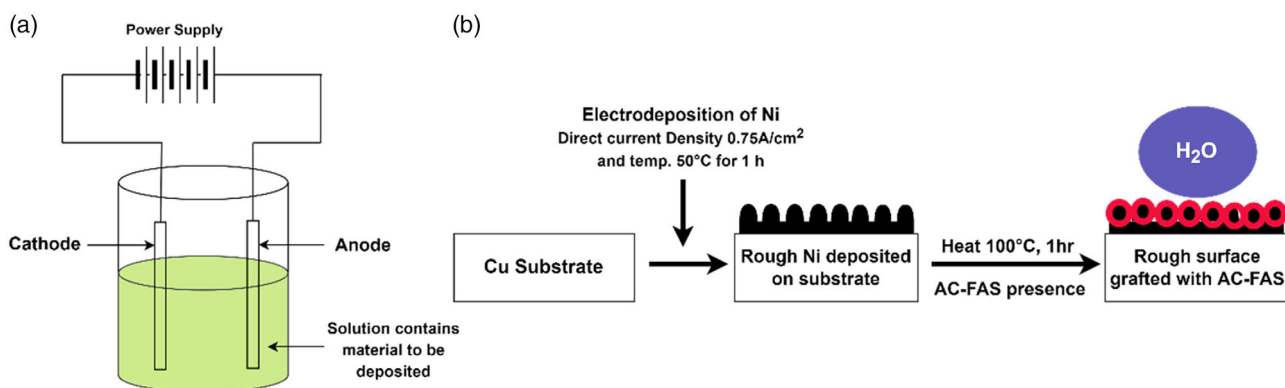


Figure 11. a) Schematic of electrodeposition setup; b) the roughness formation via electrodeposition and grafting of AC-FAS using heat treatment.^[10]

effectiveness of cathodic current and the capacity to distribute particles in the electrolyte are two main factors that influence electrodeposition.^[57]

4.2. Electrospinning Method

Electrospinning is an easy but versatile process for the growth of continuous fibers with diameters in the scale ranging from nanometer to micrometer. An electrical bias and a grounded collection plate are applied from the extrusion nozzle in the expulsion operation. Research studies have reported application of electromagnetic spinning method to produce superhydrophobic membranes with contact angles greater than 150°. The electrospay method makes it simple to combine low surface energy materials with surface roughness in one step, eliminating the need for posttreatment.^[61] Researcher reported that electrospinning method is a powerful technique to fabricate ultrafine fibers and to provide sufficient roughness and superhydrophilicity.^[62] Metals, ceramics, and polymers can all be used using this technology.^[61] In a laboratory, electrospinning requires a syringe pump, a high-power source, and a conductive substrate. The strong electric field is applied between the syringe and substrate, which initiates the process.^[63] Daewoo et al.^[63] observed this method to be versatile to produce micro/nanofibers from different polymers. The high electrical potential caused a charged liquid jet to be discharged from the tip of a distorted droplet known as a Taylor cone. Beyond a given distance, the liquid jet experiences instabilities such as whipping and bending, resulting in the formation of a solid nonwoven micro/nanofiber membrane on the substrate. By modifying the ground electrode design and/or rotating it, as well as using a microfluidic chip to feed the solution to the ejection tip, oriented polymer nanofibers can be created.^[63] Examples of superhydrophobic coating by electrospinning deposition are given in Table 3. V. Anand Ganesh et al. reported superhydrophobic coating of silica hybrid network on glass substrate by electrospinning method. The schematic of fabrication process is redrawn and shown in Figure 12. The WCA for coating was reported to be 161°. ^[64]

4.3. Wet Chemical Reaction Method

This approach encompasses a wide variety of chemical processes that directly produce the microstructures necessary for superhydrophobic surfaces. Hydrophobic components can be added during the synthesis process or afterward as posttreatments.^[30] Xuedong et al. reported the procedure to fabricate superhydrophobic coating employed in their research in which a cleaned substrate (e.g., glass or silicon wafer) was dipped into a solution of electrolyte. The solution is rapidly heated at facile temperature and coating microstructure surface was formed. Then the rough surface as prepared was modified by organic self-assembled monolayers.^[68] Researcher also reported that intense and spherical micropits appeared on the substrate after etching technique with concentrated acid (e.g., sulphuric acid, nitric acid). With the aid of ultrasonic, a superhydrophobic rough structure was prepared on the substrate material using HNO₃ etching technique. After modification with fluorinated silane, the surface showed stable super hydrophobicity.^[37–46] Dawei et al. reported that the usage of pricey low surface energy materials was the most noteworthy constraint associated with wet chemical reactions (as well as many other manufacturing methods for superhydrophobic surfaces).^[30] Examples of superhydrophobic coating by wet chemical reaction are given in Table 4. Wang et al. fabricated layer of lamellar superhydrophobic coating on wood surface by wet chemical method, with WCA of 150° and SA less than 5°. ^[69] The schematic of wet chemical method for fabricating superhydrophobic coating on wood surface is shown in Figure 13. ^[69] Hooda et al. reported that it is one of the most often used industrial procedures because it enables for the large-scale fabrication of ordered superhydrophobic coatings utilizing various materials.^[2]

4.4. Hydrothermal Synthesis Method

Hydrothermal synthesis is an effective approach to produce nanomaterials having distinctive and fascinating morphologies. Superhydrophobic nanostructure films have been successfully

Table 3. Examples of superhydrophobic coatings by electrospinning deposition method. E-PH, polyvinylidene fluoride-co-hexafluoropropylene; PVA, polyvinyl chloride; DMAc, dimethyl acetamide; DMF, dimethyl formamide; THF, tetrahydrofuran.

Matrix	Coatings	Chemical composition	Process conditions	WCA	SA/CAH	References
Al	PVDF-ZnO	PVDF + DMAc + Toluene + ZnO	Spray unit 15 kV, flow rate 1.5 mL h ⁻¹ , temp 35 °C, needle inner diameter 0.25 mm	155° ± 2°	4.5° ± 2°	[62]
E-PH	PDMS/ PVDF	PVDF + PDMS + DMF + THF	18 kV, flow rate 1.5 mL h ⁻¹ , nozzle to collector drum distance 8 cm, dried at 60 °C for more than 1 day	170°	—	[65]
Stainless steel (304 L SS)	PVA-silica	PVA + tetraethyl orthosilicate	15 kV, 2 h, flow rate 0.2 mL h ⁻¹ , nozzle to collector drum distance 18 cm,	169.2 ± 2.1°	≈3°	[66]
Si or Al	PVDF	PVDF + DMAc	Syringe tip 24 gauge metallic needle, substrate at depth ≈2 cm in liquid nitrogen, needle 10 cm above liquid nitrogen	135°	—	[67]

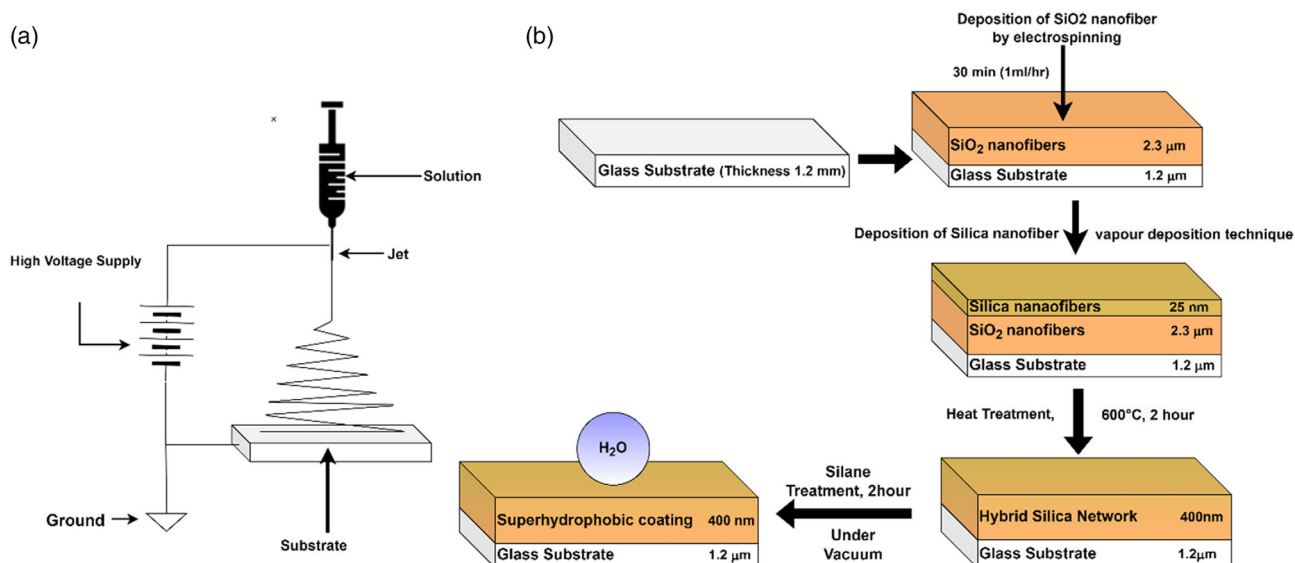


Figure 12. a) Schematic of electrospinning deposition setup; b) schematic of fabrication process of superhydrophobic hybrid silica network on glass substrate.^[64]

Table 4. Examples of superhydrophobic coatings by wet chemical reaction. FDTS, 1 H,1 H,2 H,2 H-perfluorodecyltrichlorosilane.

Matrix	Coatings	Chemical composition	Process conditions	WCA	SA/CAH	References
Si	Zn	$\text{Zn}(\text{NO}_3)_2 + \text{NH}_4\text{Cl} + \text{NH}_3 + \text{urea}$	1 hr heating at 90 °C ($10^\circ\text{C min}^{-1}$), dried substrate treated with C18	152°	–	[68]
Cotton fabric	Zn	$\text{Zn}(\text{OH})_2 + \text{ethanol} +$	Suspension ultrasonicated for 30 min, cotton fabric padded in colloidal suspension by glass rod for 10 min, fabric dried at 60 °C for 1 hr, immersed in stearic acid for 10 min, and annealed	151°	–	[70]
Mg	–	$\text{H}_2\text{SO}_4 + \text{H}_2\text{O}_2$	After etching Mg substrate is immersed in stearic acid and ethanol solution	154°	3°	[71]
Glass	GMB + Al_2O_3	Epidian + $\text{Al}_2\text{O}_3 + \text{GMB} + \text{IDA}$	Degassed composite cast on glass, allowed cross-linking 7 days at room temp. Coatings sandblasted using corundum particles, ultrasonically cleaned, RF plasma cleaning 10 min, modified with FDTS	152°	8°	[72]
Cu	Au	$\text{NaOH} + \text{Na}_2\text{S}_2\text{O}_8$	After etching Au is coated on Cu film, surface modified by <i>n</i> -dodecane thiol	151°	–	[73,74]

prepared by a facile hydrothermal method. Quick change in reversible wettability between superhydrophilicity and superhydrophobicity was achieved by regulating the adsorption/desorption process on nanostructured surfaces. Fabrication of crystalline substance under high pressure from heated aqueous solution is the main goal of this procedure. This approach is extensively used to induce roughness on the surface of a substrate by applying high pressure and temperature. Crystals are developed in autoclave, which was supplied with ingredients as well as water during the process.^[74] Dongmian et al. fabricated a lotus seedpod bioinspired AZ91D Mg alloy superhydrophobic surface via in situ hydrothermal synthesis, sonication-assisted electroless plating, and a self-assembling process. The densely packed Cu-thiolate layers on the Mg alloy substrate are uniformly

plate on the Mg alloy substrate by the predesigned TiO₂ film formed on AZ91D. The superhydrophobic coating (with self-assembled monolayer of *n*-dodecanethiol) showed WCA $\approx 158.7^\circ \pm 4.3^\circ$ and SA less than 1° .^[75] Dongmian et al. reported that superhydrophilicity and superhydrophobicity can be controlled by removing hydrophobic material *n*-dodecanthiol at elevated temperature (350 °C).^[75] Jing Yuan et al. used hydrothermal method to fabricate the superhydrophobic film by plating Ni-P on AZ61 alloy. The film showed maximum WCA of $155.6^\circ \pm 0.3^\circ$ and SA of $\approx 2^\circ$ under conditions of reaction temperature 120 °C and reaction time of 15 h.^[76] The flow chart of the fabrication process presented by Jing Yuan et al. is shown in Figure 14.^[76] Examples of superhydrophobic coatings by hydrothermal method are given in Table 5.

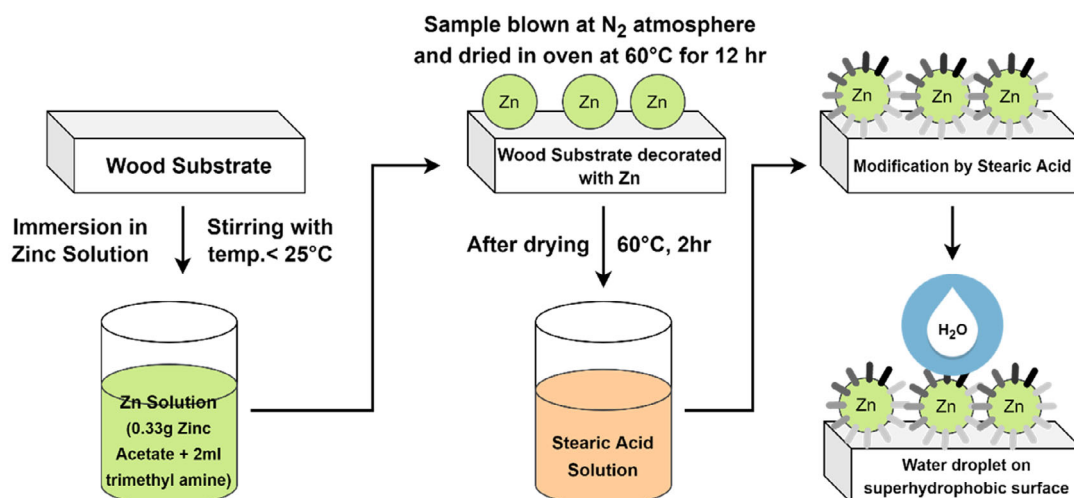


Figure 13. Schematic of wet chemical method for superhydrophobic coating on wood.^[69]

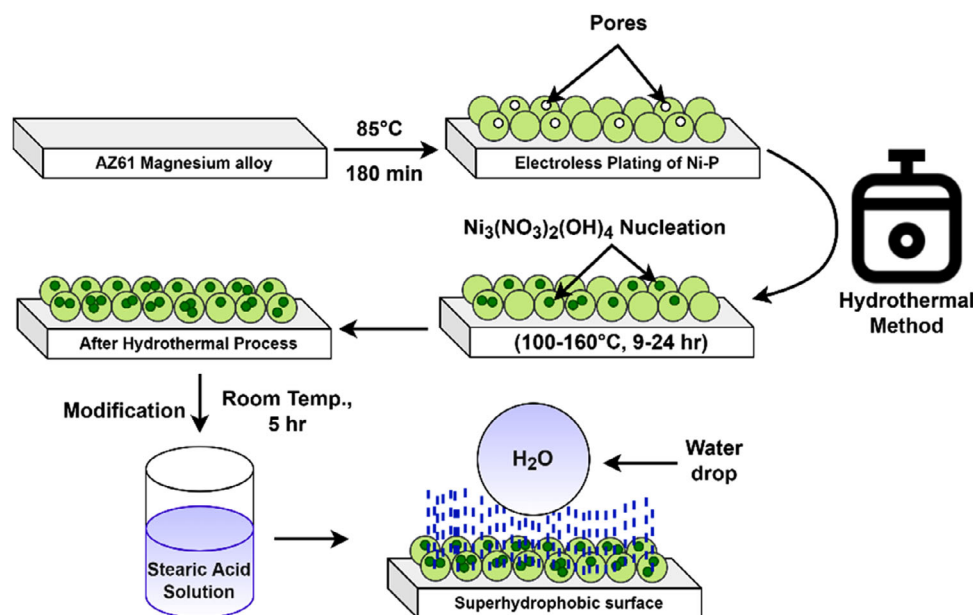


Figure 14. Schematic of hydrothermal method for superhydrophobic coating on AZ61 alloy.^[76]

4.5. Dip Coating Method

This approach is used to make nanometric thick films. In this method, the substrate is submerged in a solution of the material to be deposited. Thickness of the film is controlled by pulling rate of substrate from solution. The higher is the rate, the thinner is the film. High output, uniform, very durable, compact, stable coating, easily repairable, and simultaneous coating on the top and bottom parts of the substrate are some of the advantages. However, all components used in the coatings have to be submersible.^[74] Xiaoli Liu et al. reported the synthesis of superhydrophobic polydimethylsiloxane/stearic acid/silica fabric. This superhydrophobic fabric possesses WCA of 163° and this fabric

efficiently repelled the liquids like Coca-Cola, milk, coffee, orange juice, 1M NaOH, and HCl.^[82] Nurul F. Himma et al. fabricated the superhydrophobic PP hollow fiber membrane by dip coating method. This fabricated membrane has WCA of 151.3°. Nurul F. Himma et al. reported that WCA slightly declined with increase in number of dipping cycle. Further, author concluded that the increasing the number of dipping cycle, increase in the coating thickness, and decrease in the surface roughness resulted in declination of WCA. Increasing withdrawal velocity results in increase of WCA until reached the peak withdrawal velocity of 15 mm s⁻¹ and further increase in withdrawal velocity decreases the WCA.^[83] Meikandan et al. developed the superhydrophobic material by coating AgNO₃ on Cu substrate by using

Table 5. Examples of superhydrophobic coatings by hydrothermal method.

Substrate	Coating	Chemical composition	Process conditions	WCA	SA/CAH	References
Al	$\text{Al}[\text{CF}_3(\text{CF}_2)_{12}\text{COO}]_3$	$\text{Al}_2\text{O}_3 + \text{F}_3(\text{CF}_2)_{12}\text{COOH} + \text{ethanol}$	150 °C, 1 h	$158^\circ \pm 2^\circ$	$3.5^\circ \pm 0.5^\circ$	[77]
Mg	$\text{Zn}[\text{CH}_3(\text{CH}_2)_{16}\text{COO}]_2$	$\text{ZnCl}_2 + \text{H}_2\text{O} + \text{Stearic acid} + \text{ethanol}$	80 °C, 2 h	$167^\circ \pm 0.8^\circ$	$4.8^\circ \pm 0.9^\circ$	[78]
Mg	$\text{Ce}[\text{CH}_3(\text{CH}_2)_{16}\text{COO}]_2/\text{CeO}_2$	Stearic acid + ethanol + Cerium oxide (CeO_2) +	170 °C, 10 h	164°	4.1°	[79]
Cu	$2\text{Cu}[\text{CH}_3(\text{CH}_2)_{16}\text{COO}]$	Ethanol + stearic acid	Etched Cu (in NH_3) kept in autoclave with NaOH for 2 h at 200 °C, finally Cu immersed in stearic acid solution	$157.7^\circ \pm 1^\circ$	–	[80]
Glass slide (soda lime glass)	–	POTS	120 °C, 2 h, The vapor of POTS reacts with substrate. Finally, slide kept in oven at 150 °C for 1.5 h	160°	1°	[81]

simple dip coating method. The developed superhydrophobic coating showed WCA of 159°. The fabric retained WCA > 150° even after 72 h UV radiation and 700 cycles of mechanical abrasion.^[84] The process schematically represented by M. Meikandan et al. for superhydrophobic coating by dip coating is shown in Figure 15.^[84] Examples of superhydrophobic coatings by dip coating method are given in Table 6.

4.6. Self-Assembly and Layer-by-Layer Methods

Eduardo Guzman et al. reported that two types of assemblies dominated the fabrication of soft nanomaterials: 1) self-assembled monolayers (SAM) and 2) Langmuir–Blodgett (LB) and Langmuir–Schaeffer (LS). SAMs are organized molecule assemblies that are adsorbed onto surfaces and formed via a chemical interaction, such as silanes on silicon surfaces or thiols on gold. The formation of LB and LS films, on the other hand, is accomplished by transferring amphiphilic molecules from a water/vapor interface to a solid surface via a mechanism that allows for the deposition of several layers.^[90]

Self-assembly technique is useful to get the organized system from components of preexisting disordered system. The layers of coating are grown at low temperature and with minimal

intervention.^[74] Multilayer grafts are formed by electrostatic interactions and covalent bonds. This method is simple and the most popular for the deposition of superhydrophobic coatings. It can be employed for the deposition of various types of materials polymers, metals, ceramics, nanoparticles, and biological molecules.^[74] Researcher reported that superhydrophobic surfaces were prepared by multilayer layer assembly of polyelectrolyte/silica nanoparticles followed by a fluoroalkyl silane treatment through electrostatic layer. The surface morphology of the coated surface of silica nanoparticles resulted in the hydrophobicity range, and can be tailored by regulating the number of multilayers.^[48,49]

The SAMs, LB, and LS techniques, on the other hand, are limited to the production of coatings on flat macroscopic surfaces, requiring extraordinarily clean substrates and a dust-free atmosphere for successful film deposition. LB and LS films also have limited mechanical and thermal resistance, and their manufacture is time-consuming. The constraints of SAMs, LB, and LS films were somewhat solved with the creation of the layer-by-layer (LbL) technique.^[90]

Because the LbL technique was developed for the deposition of multilayered coatings onto flat surfaces, it should be considered a template-assisted assembly process. As a result, a precursor

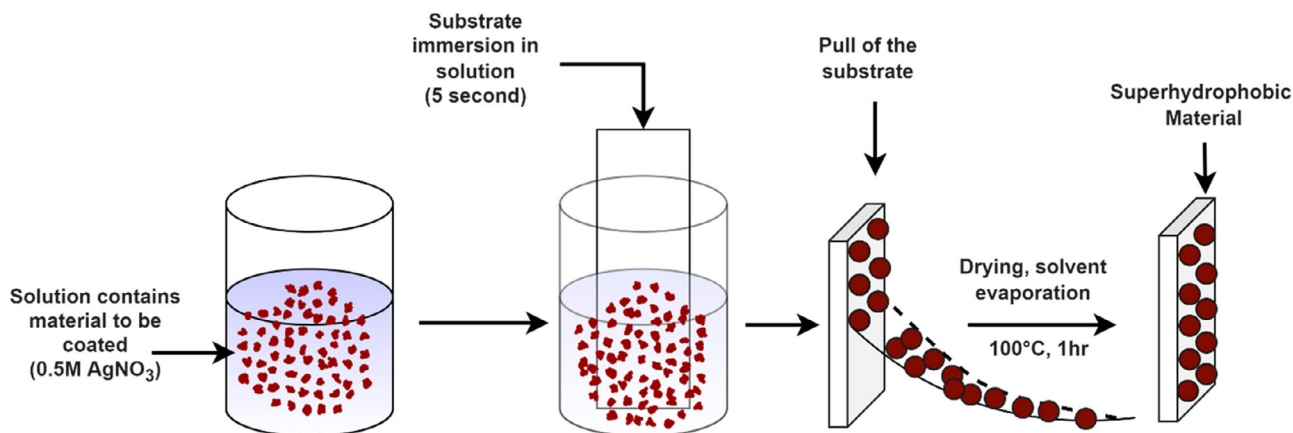


Figure 15. Schematic of dip coating method for superhydrophobic coating on Cu substrate.^[84]

Table 6. Examples of superhydrophobic coatings by dip coating method.

Substrate	Coating	Chemical composition	Process conditions	WCA	SA/CAH	References
Quartz	Silica	Methyltrimethoxysiloxane + methanol + hexane + oxalic acid	Dip coating through double step sol-gel. Substrate dipped in gel for 10 hr at room temp. Annealing at 150 °C, 2 h	168° ± 2°	3° ± 1°	[85]
Cu plate, Si wafer, and glass slip.	PLLA/SiO ₂	PLLA + SiO ₂	Dip for 30 s in composite, dried for 20 min at room temp.	155.7° ± 1.4°	5.2° ± 0.4°	[86]
Glass slide	Si	Methyltrichlorosilane + pristine SiO ₂ + toluene	Dip for 5 s at room temp., withdrawn rate 5 mm s ⁻¹ , kept in air 5 min	153° ± 2°	8° ± 1°	[87]
PP hollow fiber	PP	PP + xylene	Dip for 30 s in nonsolvent (MEK) followed by 10 s in PP solution. Withdrawal speed 8-25 mm s ⁻¹ . Sample dried for 1 h at const. temp.	149° ± 2.3°	–	[88]
Wool fabric	SiO ₂ /oligomer	Ethanol + water + NH ₃ + tetraethoxysilane + toluene	Dip in composite and ultrasonicated for 0.5 h. Finally cured at 60 °C for 50 min.	–	3°	[89]

substrate, whether sacrificial or not, is required for the fabrication of LbL films to support the layer assembly. Nowadays, any solvent-accessible surface, independent of chemical origin, shape, geometry, or size, may be employed as a template for the production of LbL materials. Some of the current LbL coating templates include colloidal micro- and nanoparticles, liposomes or vesicles, micelles, fluid interfaces (floating multilayers), emulsion droplets, or even cells.^[90] LbL methods have different approach for deposition like dip coating, spin coating, and spray coating assisted LbL approach. Schematic of these approach is redrawn and shown in **Figure 16**.^[90] Huaiyuan et al.^[91] synthesized superhydrophobic coating on potassium titanate whiskers by polytetrafluoroethylene/polyetheretherketone composite with self-assembled octadecyl-trichlorosilane. Water contact angle for the coating was reported as 141°. Examples of superhydrophobic coating by LbL method are given in **Table 7**.

4.7. Colloidal Templatation Method

This method introduces a predetermined structure (template) to the self-assembly. Colloidal are aggregated with controlled shape, size, and structures by removing solvents from aqueous building blocks placed on a template's patterned surface. This technique combines the advantages of self-assembly with lithography. At the nanoscale level, template-assisted self-assembly also allows for greater flexibility in ordering the building blocks.^[74] Pernites et al.^[97] reported the two-step approach for fabrication of antiwetting surface. Author started with colloidal template (polystyrene particle) layering on planar conducting substrate (Au) by LB method. Further polythiophene is deposited on PS-modified substrate. This fabricated superhydrophobic coating showed WCA of 154° ± 1°. Yue Li et al.^[98] reported the synthesis of superhydrophobic coating by colloidal template method. The schematic of method is shown in **Figure 17**. Examples of superhydrophobic coating by dip coating method are given in **Table 8**.

4.8. Sol-Gel Method

This method is mainly used for biotemplating or bioinspired fabrication of superhydrophobic coatings. In sol-gel approach,

colloidal particles (nanometer range) are dispersed in gel, which forms interconnected network. The conversion of monomer into the colloidal solution acts as an initiator for gel formation.^[74] A simple sol-gel dip-coating process based on reversible

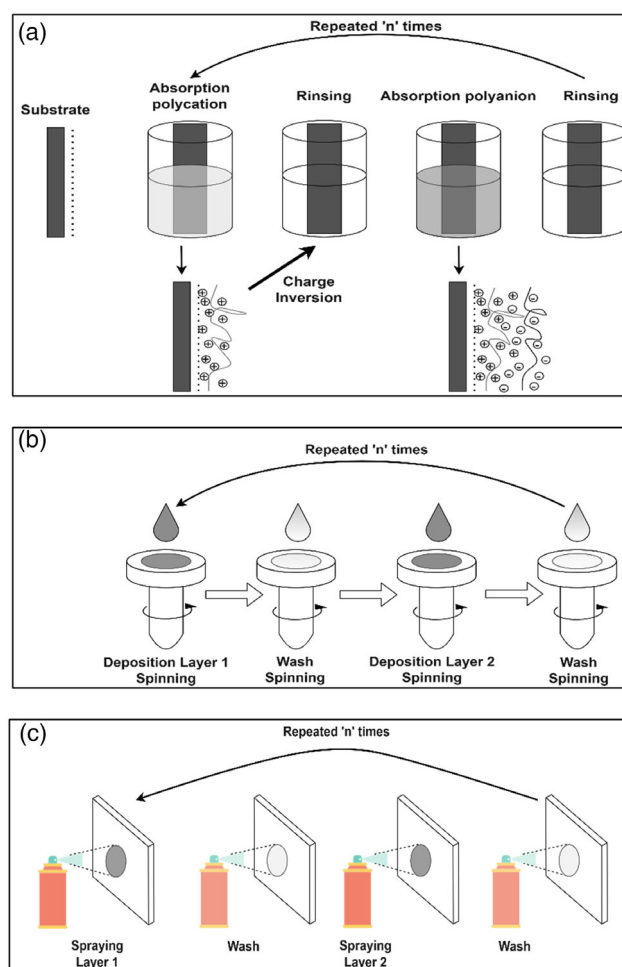


Figure 16. Schematic of LbL method for superhydrophobic coating assisted with a) dip coating, b) spin coating, and c) spray coating.^[90]

Table 7. Examples of superhydrophobic coatings by LbL method. PAH, poly(allylamine hydrochloride); PAA, poly(acrylic acid); PSC, polyaniline silica composite; PDDA, polydiallyldimethylammonium chloride; PSP, poly(sodium phosphate); PSS, poly 4-styrene sulphonate.

Substrate	Coating	Chemical composition	Process conditions	WCA	SA/CAH	References
Cotton fabric	Polyelectrolyte/silica	1) PAA + NaH ₂ PO ₂ 2) PAH 3) PAA 4) SiO ₂ solution	Electrostatic assembly, pH of PAH and PAA adjusted by NaOH to 7.5 and 7, immersion time for each layer is 5 min followed by rinsing.	>150°	<10°	[92]
316SS	PSC/TMS-SiO ₂	1) PSC 2) TMS-SiO ₂	Spin coating assembly, layer of solution 1 and 2 dropped on substrate for 30 s at 500 rpm accelerated to 1000 rpm for 1 min. Deposited film heated at 110° for 30 s	153° ± 2°	6° ± 2°	[93]
Quartz	(PDDA-Silicate/PAA)/ (PAH-SiO ₂)	1) PAH 2) SiO ₂ solution 3) PDPA	(PDDA-Silicate/PAA) 12 under layer and (PAH-SiO ₂) 4 top layer, calcined at 500 °C	157°	2°	[94]
Glass	[PSP/PAH] ₄₀	1) PAH 2) PSP	CVD approach used for deposition by LbL method. For both solution conc. is 10 ⁻³ M. Ionic strength of mixture set using 0.15 M NaCl. pH adjusted to 6.4 by adding HCl or NaOH.	161°	—	[95]
Indium tin oxide (ITO)	Ag-PDDA/PSS	1) PDPA 2) PSS 3) AgNO ₃ + NaNO ₃	Dip time of ITO electrode in PDPA and PSS solution 5 min each until desired number of layers formed. Ag aggregates deposited on PDPA/PSS modified substrate by electrodeposition	154°	3°	[96]

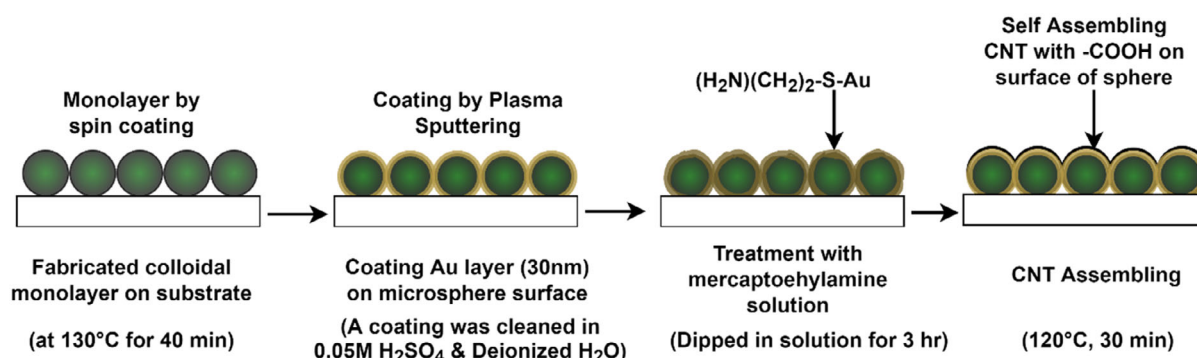


Figure 17. Schematic of colloidal templation method for superhydrophobic coating.^[98]

aggregation of silica colloidal particles can be used to make an organic–inorganic composite coating that is superhydrophobic, high visibility, and thermally stable. Particles of the silica colloid aggregated with poly-aminopropyltriethoxysilane (poly-APS) were partially redispersed by ultrasonication, creating several small aggregates. Consequently, these aggregates were coated with dip coating. The APS concentration can be modified to regulate surface structure, superhydrophobicity, and transparency.^[50–53] The sol–gel technique does not require high temperatures or pressures, and it can be used on a wide variety of substrates.^[26]

Xiu et al.^[103] report that sol–gel method can be used to make porous glass films in a controlled environment. The moderate preparation conditions allow a wide spectrum of labile organic species to be included into a glass composite. Sol–gel generated materials have variable porosity, transparency, hardness, and high thermal stability, in addition to the gentle manufacturing conditions. Author also reports that UV stability is a prerequisite

for outdoor applications. Yanping Zhang et al.^[104] fabricated superhydrophobic silica film by sol–gel method. The fabricated film modified with trimethylchlorosilane (TMCS) shows WCA of 164° ± 2°. The schematic of the process is redrawn and presented in **Figure 18**. Examples of superhydrophobic coating by sol–gel method are given in **Table 9**.

5. Future Scope

Future research on scalability, durability, and maintenance of superhydrophobic coatings is required. Fabrication of environmentally friendly superhydrophobic polymer nanocomposites which create less unstable organic emissions is also attracting the attention of researchers. There is a strong need to explore more economic fabrication techniques for large-scale production of durable, environment-friendly superhydrophobic coatings for a variety of applications.

Table 8. Examples of superhydrophobic coatings by colloidal templation method. HDPE, high-density polyethylene; PFPE, perfluoropolyether; SS, fluorinated styrene sulfonate ester.

Substrate	Coating	Chemical composition	Process conditions	WCA	SA/CAH	References
Glass	CNT/Au-PS	1) PS solution 2) H ₂ SO ₄ 3) CNT solution 4) Mercaptoethylamine solution	PS coated on substrate by spin coating and then heated at 130° for 40 min. By plasma sputtering gold layer deposited on surface. After treatment with mercaptoethylamine solution CNT solution is dropped on the surface of colloidal crystal.	165°	SA-5°	[98]
Glass	ZnO	1) PS solution 2) Zinc acetate	PS colloidal crystal coated by spin coating and further 2D ordered porous ZnO structure fabricated by dip coating.	165°	SA-<5°	[99]
Taro leaf (original template), PDMS (negative template)	PS	1) PS solution 2) PDMS solution	PDMS solution casted on Taro leaf, after solidification (24 h), PDMS film is peeled off and used as negative template. PS solution cast on negative template and after solvent vaporization (10 h) PS film is peeled off.	158°±1.6°	SA-3°	[100]
<i>p</i> -AAO (template)	PFPE-SS	1) Phosphoric acid 2) PFPE + SS	Al film is anodized at 180 V at 2 °C. With controlled anodization time <i>p</i> -AAO formed. PFPE + SS cured on <i>p</i> -AAO surface and peeled off.	171°	SA-3°	[101]
Etched Al (template)	HDPE	1) HCL + HF + H ₂ O 2) HDPE plate	Etched Al template is replicated on HDPE by heat and pressure treatment.	160°	CAH-2°	[102]

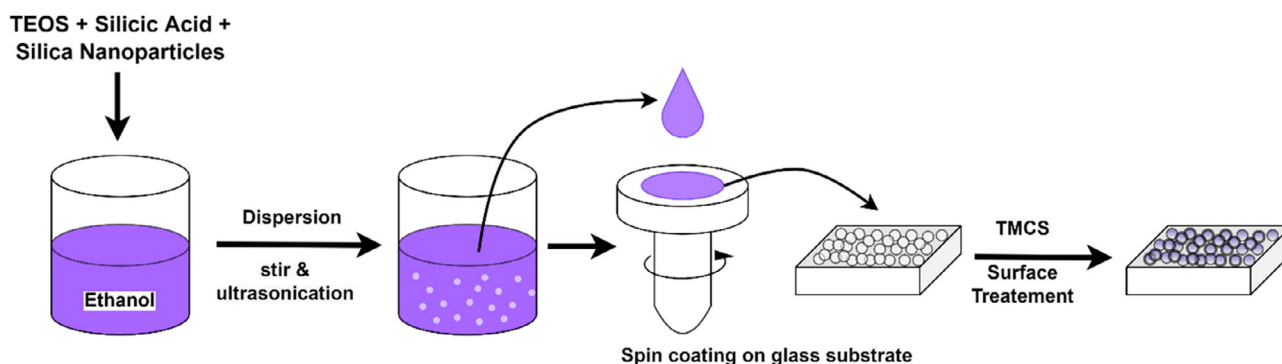


Figure 18. Schematic of sol-gel method for superhydrophobic coating.^[104]

The most promising manufacturing technique in the future will be biologically inspired microstructure surface fabrication. Preparation methods are becoming more diversified and understandable as a result of the rapid development of several new manufacturing technologies and bionic engineering, and environmentally benign materials are being employed to prepare bionic materials. Composite drag reduction technology is expected to become a focus of study in the coming decade. Biologically activated superhydrophobic materials for reducing underwater drag have a promising future, particularly in commercial manufacture.^[108]

Making durable, long-lasting coatings without employing any “wet” formulas or dispersions is probably the most difficult element. To create superhydrophobic coatings, some potential solvent-free approaches have been devised, but these either required the use of fluorinated polymer nanoparticles or specially manufactured or designed substrates. In this case, the concept of triboelectric adhesion or coating, which has recently been developed as nanogenerators, could be a potential strategy. It is

expected that superhydrophobic coatings based on tribocharging of particles can be created without the use of solvents or wet chemical procedures with some changes. The demand for green manufacturing technologies has grown as a result of severe environmental restrictions, government legislation, and growing customer demand for safer and healthier coating formulas.^[1]

6. Conclusion

The present work reviews the recent development, fabrication, and applications of superhydrophobic polymer nanocoatings. The review suggests that addition of nanoparticles increases the superhydrophobicity of polymer matrices. The present work highlights the different manufacturing methods and techniques used in fabrication of superhydrophobic coatings. Along with the manufacturing steps involved, the processing and application are also discussed. Currently, the big challenge is to have coatings that are easy to apply and have long-term stability alongside

Table 9. Examples of superhydrophobic coatings by sol–gel method. PET, poly(ethylene terephthalate); FAS, fluoroalkyl silanes; FLKS10, α,ω -triethoxysilane terminated perfluoropolyether; TEOT, tetraethyl orthotitanate; 17FTMS, (heptadecafluoro-1,1,2,2-tetrahydrodecyl)trimethoxysilane; MTES, methyltriethoxysilane.

Substrate	Coating	Chemical composition	Process conditions	WCA	SA/CHA	References
Glass	Silica	1) Tetramethoxysilane 2) Isobutyltrimethoxysilane	Silica film formed on glass by casting. Polybutadiene film is cast on as prepared silica film by spin coating and subsequently surface is exposed to SF ₆ . This form superhydrophobic fluorinated polybutadiene film.	160°	CHA < 2°	[103]
Glass	TiO ₂ –ZrO ₂ –SiO ₂	1) FLKS10 2) TEOS 3) TEOT 4) TPOZ 5) THF	TiO ₂ –ZrO ₂ –SiO ₂ and silane terminated PFPE coating on glass substrate fabricated by combining air brushing and spin coating. Multilayer sample with 2–5 layers shows high value of surface roughness.	157°	CHA < 2°	[105]
Glass	Long-chain FAS	17FTMS	Ammonia used as catalyst to initiate sol–gel process. Water used to accelerate hydrolysis reaction. The coating was formed by dip coating.	169°	5°	[106]
PET	Al ₂ O ₃	1) Al(O-sec-C ₄ H ₉) ₃ + IPA 2) EAcAc	Dip coting technique is used with withdrawal speed 2 mm s ^{−1} . Al ₂ O ₃ film dried at room temp. and then immersed in hot water (60 °C)	>150°	–	[107]
Glass	Organically modified silica	1) MTES 2) TMCS	Hydrophobic alcohol prepared and sprayed on hot glass substrate at 100 °C. After, the coating surface is modified with TMCS.	167°±1°	2°±1°	

the technology that allows for combinations of the properties of different materials that are otherwise difficult.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

contact angle, nanomaterials, superhydrophobic coatings, surface tension, wettability

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